

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name Nitrate 3 (as NO₃)-HR

Product No AC4007-FOIL

Pure substance/mixture Mixture

Contains Cadmium

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

Supplier

Fisher Scientific Korea	Thermo Fisher Scientific©
D5,D6, Incheon Airport Logistics	Water and Lab Products
Complex	22 Alpha Road
150, Gonghangdong-Ro Mapo-Gu 296	Chelmsford, MA 01824, USA
Beon-Gil	1-978-232-6000
Jung-Gu, Incheon 400-340	
Tel: +82-2-6196-5500	
Fax: +82-2-6196-5501	

E-mail address

Chem.KR@thermofisher.com info.water@thermo.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

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Health hazards

Acute Inhalation Toxicity - Dusts and Mists	Category 3
Serious Eye Damage/Eye Irritation	Category 1
Germ Cell Mutagenicity	Category 2
Carcinogenicity	Category 1
Reproductive Toxicity	Category 2
Specific target organ toxicity - (repeated exposure)	Category 2

Environmental hazards

Chronic aquatic toxicity	Category 2
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Label Elements



Signal Word

Danger

Hazard Statements

H341 - Suspected of causing genetic defects
H350 - May cause cancer
H361 - Suspected of damaging fertility or the unborn child
H411 - Toxic to aquatic life with long lasting effects
H318 - Causes serious eye damage
H331 - Toxic if inhaled
H341 - Suspected of causing genetic defects if inhaled
H373 - May cause damage to organs through prolonged or repeated exposure

Precautionary Statements

Prevention

P201 - Obtain special instructions before use
P260 - Do not breathe dust/fume/gas/mist/vapors/spray
P271 - Use only outdoors or in a well-ventilated area
P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection
P261 - Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray
P281 - Use personal protective equipment as required
P273 - Avoid release to the environment

Response

P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/ physician
P311 - Call a POISON CENTER or doctor/ physician
P321 - Specific treatment (see supplemental first aid instructions on this label)
P308 + P313 - IF exposed or concerned: Get medical advice/ attention
P314 - Get medical advice/ attention if you feel unwell
P391 - Collect spillage

Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P405 - Store locked up

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

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Other Hazards

NFPA

Health
3

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Tartaric Acid	No information available	87-69-4	KE-10801	20 - 30
Sodium Sulfate	No information available	7757-82-6	KE-31609	20 - 30
Sodium Tartrate	No information available	868-18-8	KE-12382	>=10 - 20
CDTA Trisodium Salt	No information available	36679-96-6	Not listed	>=10 - 20
Magnesium Sulfate, Anhydrous	No information available	7487-88-9	KE-22752	>=5 - <10
Cadmium	No information available	7440-43-9	KE-04397	>=5 - <10

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation

Move to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

Ingestion

Do not induce vomiting. Call a physician or Poison Control Center immediately.

Self-Protection of the First Aider

Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects

Causes eye burns Causes severe eye damage

Indication of any immediate medical attention and special treatment needed

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Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

Environmental Precautions

Do not flush into surface water or sanitary sewer system.

Methods and Material for Containment and Cleaning Up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment
Do not get in eyes, on skin, or on clothing
Avoid dust formation
Use only under a chemical fume hood
Do not breathe vapors/dust
Do not ingest

Conditions for Safe Storage, Including any Incompatibilities

Storage

Keep container tightly closed in a dry and well-ventilated place
Store at room temperature in the original container
Keep away from direct sunlight

Incompatible Products

No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Tartaric Acid	87-69-4	Not listed	Not listed	Not listed
Sodium Sulfate	7757-82-6	Not listed	Not listed	Not listed

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Sodium Tartrate	868-18-8	Not listed	Not listed	Not listed
CDTA Trisodium Salt	36679-96-6	Not listed	Not listed	Not listed
Magnesium Sulfate, Anhydrous	7487-88-9	Not listed	Not listed	Not listed
Cadmium	7440-43-9	TWA: 0.01 mg/m ³ TWA: 0.002 mg/m ³	TWA: 0.01 mg/m ³ TWA: 0.002 mg/m ³	Ceiling: 0.3 mg/m ³ Ceiling: 0.6 mg/m ³ (Vacated) STEL: 0.3 ppm TWA: 0.1 mg/m ³ TWA: 0.2 mg/m ³ TWA: 5 µg/m ³

Component	CAS-No	European Union	The United Kingdom	Germany
Tartaric Acid	87-69-4	Not listed	Not listed	TWA: 2 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 mg/m ³
Sodium Sulfate	7757-82-6	Not listed	Not listed	Not listed
Sodium Tartrate	868-18-8	Not listed	Not listed	Not listed
CDTA Trisodium Salt	36679-96-6	Not listed	Not listed	Not listed
Magnesium Sulfate, Anhydrous	7487-88-9	Not listed	Not listed	Not listed
Cadmium	7440-43-9	Not listed	STEL: 0.075 mg/m ³ 15 min TWA: 0.025 mg/m ³ 8 hr Carc. metal	Haut

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Tartaric Acid	87-69-4	Not listed
Sodium Sulfate	7757-82-6	Not listed
Sodium Tartrate	868-18-8	Not listed
CDTA Trisodium Salt	36679-96-6	Not listed
Magnesium Sulfate, Anhydrous	7487-88-9	Not listed
Cadmium	7440-43-9	5 µg/g creatinine Medium: urine Time: not critical Determinant: Cadmium 5 µg/L Medium: blood Time: not critical Determinant: Cadmium

Appropriate engineering controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:.
Face-shield.

Skin and Body Protection Wear protective gloves/clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

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Environmental exposure controls Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Solid
Appearance	Blue Metallic Powder
Odor	Odorless
Odor Threshold	No information available
pH	3.4
PH Range	1.9-4.9

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	No information available	
Flash Point (High in °C)	N/A	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

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No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	Not an expected route of exposure.
Ingestion	May be harmful if swallowed.
Eyes	Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness.
Skin	Avoid contact with skin. Skin corrosion/irritation.

Information on Toxicological Effects

Symptoms No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Sensitization No information available

Mutagenic Effects No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	ACGIH	IARC	NTP	OSHA
Cadmium 7440-43-9	A2	Group 1	Known	X

ACGIH: (American Conference of Governmental Industrial Hygienists)

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

IARC: (International Agency for Research on Cancer)

IARC: (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program)

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

Reproductive Effects No information available

STOT - single exposure No information available

STOT - repeated exposure No information available

Aspiration hazard No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 3581 mg/kg

ATEmix (dermal) 4767 mg/kg

ATEmix (inhalation-dust/mist) 6.5 mg/L

SECTION 12: ECOLOGICAL INFORMATION

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Ecotoxicity

60% of the mixture consists of component(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Sodium Sulfate 7757-82-6	-	LC50: 3040 - 4380 mg/L, 96h static (Lepomis macrochirus) LC50: = 13500 mg/L, 96h (Lepomis macrochirus) LC50: > 6800 mg/L, 96h static (Pimephales promelas) LC50: 13500 - 14500 mg/L, 96h (Pimephales promelas)	EC50: = 2564 mg/L, 48h (Daphnia magna) EC50: = 630 mg/L, 96h (Daphnia magna)
Magnesium Sulfate, Anhydrous 7487-88-9	EC50: = 2700 mg/L, 72h (Desmodesmus subspicatus)	LC50: 2610 - 3080 mg/L, 96h static (Pimephales promelas) LC50: = 19000 mg/L, 24h static (Lepomis macrochirus)	EC50: = 1700 mg/L, 24h (Daphnia magna) EC50: 266.4 - 417.3 mg/L, 48h Static (Daphnia magna)
Cadmium 7440-43-9	-	LC50: 0.0004 - 0.003 mg/L, 96h (Pimephales promelas) LC50: = 0.016 mg/L, 96h (Oryzias latipes) LC50: = 21.1 mg/L, 96h flow-through (Lepomis macrochirus) LC50: = 0.24 mg/L, 96h static (Cyprinus carpio) LC50: = 4.26 mg/L, 96h semi-static (Cyprinus carpio) LC50: = 0.002 mg/L, 96h (Cyprinus carpio) LC50: = 0.006 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 0.003 mg/L, 96h flow-through (Oncorhynchus mykiss)	EC50: = 0.0244 mg/L, 48h Static (Daphnia magna)

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

Some components of this material have some potential to bioaccumulate

Mobility

No information available.

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste Disposal Methods

Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

Not Regulated

IATA

Not regulated

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IMDG/IMO	Not regulated
Marine Pollutant	Not Applicable
Special Precautions for User	No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Tartaric Acid	87-69-4	KE-10801	X	201-766-0	X	X	-	X	X	X
Sodium Sulfate	7757-82-6	KE-31609	X	231-820-9	X	X	-	X	X	X
Sodium Tartrate	868-18-8	KE-12382	X	212-773-3	X	X	-	X	X	X
CDTA Trisodium Salt	36679-96-6	-	-	-	X	X	-	-	-	-
Magnesium Sulfate, Anhydrous	7487-88-9	KE-22752	X	231-298-2	X	X	-	X	X	X
Cadmium	7440-43-9	KE-04397	X	231-152-8	X	X	-	X	-	X

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	Not applicable	Not applicable	Annex I - Y26

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Tartaric Acid	87-69-4	Listed	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Listed	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Listed	Not applicable	Not applicable
Cadmium	7440-43-9	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Tartaric Acid	87-69-4	Annex 1 - KE-10801	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Annex 1 - KE-31609	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Annex 1 - KE-12382	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Annex 1 - KE-22752	Not applicable	Not applicable
Cadmium	7440-43-9	Annex 1 - KE-04397	Not applicable	Listed

Component	CAS-No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable

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CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	97-1-250 (>25%)	Not applicable	06-5-9 (>0.1%)

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	>0.3 mg/L	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Listed	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Listed	Listed	0.01 mg/m³ TWA 0.002 mg/m³ TWA

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	TWA: 0.01 mg/m³ TWA: 0.002 mg/m³	Formulations containing 0.1% or more

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising	Class 2 - Flammable	Class 3 - Spontaneously	Class 4 - Flammable	Class 5 - Self-reactive	Class 6 - Oxidising
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		solids	solid	Combustible Substances and Dangerous Substances When Wet	liquids	substances	liquids
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	5. Metal powder 500 kg powder	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Tartaric Acid	87-69-4	Not listed	Not listed
Sodium Sulfate	7757-82-6	Not listed	Not listed
Sodium Tartrate	868-18-8	Not listed	Not listed
CDTA Trisodium Salt	36679-96-6	Not listed	Not listed
Magnesium Sulfate, Anhydrous	7487-88-9	Not listed	Not listed
Cadmium	7440-43-9	TWA: 0.01 mg/m ³ TWA: 0.002 mg/m ³	5 µg/g creatinine Medium: urine Time: not critical Determinant: Cadmium 5 µg/L Medium: blood Time: not critical Determinant: Cadmium

US Management Information

OSHA - Occupational Safety and Health Administration

OSHA - United States Occupational Safety and Health Administration

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Tartaric Acid	87-69-4	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable
Cadmium	7440-43-9	5 µg/m ³ TWA 2.5 µg/m ³ Action Level	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Tartaric Acid	87-69-4	Not applicable	Not applicable	Not applicable
Sodium Sulfate	7757-82-6	Not applicable	Not applicable	Not applicable
Sodium Tartrate	868-18-8	Not applicable	Not applicable	Not applicable
CDTA Trisodium Salt	36679-96-6	Not applicable	Not applicable	Not applicable
Magnesium Sulfate, Anhydrous	7487-88-9	Not applicable	Not applicable	Not applicable
Cadmium	7440-43-9	Not applicable	10 lb	0.1

CLP Classification - Regulation (EC) No 1272/2008

Danger.

H331 - Toxic if inhaled. H318 - Causes serious eye damage. H341 - Suspected of causing genetic defects. H350 - May cause cancer. H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child. H373 - May cause damage to organs through prolonged or repeated exposure. H411 - Toxic to aquatic life with long lasting effects.

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P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/ physician. P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection.

SECTION 16: OTHER INFORMATION

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.©
Issue Date	05-Sep-2019
Revision Date	05-Sep-2019
Reason for revision	Initial Release.

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

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